System Architecture & Component Interface Specification

Project Context: Collaborative Audio Effects Chain Synthesis via Differentiable RL & Human-in-the-Loop Personalisation
Purpose: Cross-component interface contracts, schemas, tensor formats, and PTA-to-M3 integration guidelines.

---

1. System Architecture & Status Matrix

This specification documents the current deliverables, tensor specifications, and interchange formats between **M1** (Analysis), **M2** (Inverse Processing), **M3** (RL FX Chain Generator), and **PTA** (Personalized Taste Adapter & XAI).

Module	Component Responsibility	Implementation Status	Key Deliverables & Code Pointers
M1	Audio Analysis & Feature Extraction	**IMPLEMENTED**	examples/analyze_audio.py (Pure PyTorch acoustic, temporal, dynamic, and spectral metrics extraction).
M2	Audio Understanding & Inverse FX	**IMPLEMENTED & TRAINED**	models/inverse_fx_net.py (ResNet backbone + 7-effect classifier + 29-parameter regression + DAPN core).
M3	RL-Based FX-Chain Generation	**PLANNED (Env Ready)**	dapn/dapn.py (Differentiable execution engine & MR-STFT loss ready; PPO/GAE agent loop in progress).
PTA	Personalized Taste Adapter & XAI	**IN INTEGRATION PREP**	Preference vector conditioning, reward guidance shaping, and explainable constraints for M3 synthesis.

2. M1 — Audio Analysis / Feature Extraction Interface

Component Scope: Deterministic acoustic extraction across **Dry un-processed audio (x)**, **Target Reference audio (y_ref)**, and **Generated wet audio (y_gen)**. Outputs a structured feature dictionary or PyTorch feature vectors.

- **Audio Input:** WAV tensor (C, T) or batch (B, C, T) at 44.1 kHz, single-channel mono or stereo downmixed.
- **Front-end Delta:** In addition to absolute metrics, M2 computes differential Log-Mel spectrograms: Mel_delta = Mel(wet) - Mel(dry).

Feature Metric	Class	Physical Unit	Raw Range	Normalized ML Scale [0, 1]
peak_linear / peak_dbFS	Dynamics	Linear / dBFS	[0.0, 1.0] / [-inf, 0]	(dBFS + 80.0) / 80.0
rms_linear / rms_dbFS	Dynamics	Linear / dBFS	[0.0, 1.0] / [-inf, 0]	(dBFS + 80.0) / 80.0
crest_factor_db	Dynamics	dB	[0.0, 40.0+]	crest_db / 40.0
zero_crossing_rate	Temporal	rate per sample	[0.0, 1.0]	Direct normalized [0.0, 1.0]
spectral_centroid_hz	Spectral	Hz	[20.0, 22050.0]	Log10 min-max or normalized [0, 1]
spectral_bandwidth_hz	Spectral	Hz	[0.0, 11025.0]	bandwidth / 11025.0
spectral_rolloff_85pct_hz	Spectral	Hz	[20.0, 22050.0]	rolloff / 22050.0
spectral_flatness	Spectral	Wiener ratio	[0.0, 1.0]	Direct normalized [0.0, 1.0]
low_sub_bass_under_250hz_pct	Band Energy	% of energy	[0.0, 100.0] %	pct / 100.0
midrange_250_to_4000hz_pct	Band Energy	% of energy	[0.0, 100.0] %	pct / 100.0
high_presence_above_4000hz_pct	Band Energy	% of energy	[0.0, 100.0] %	pct / 100.0

3. M2 — Audio Understanding / Inverse Processing Interface

Model Architecture (InverseFXNet): End-to-end deep convolutional neural network mapping paired dry and wet audio waveforms directly to active effect presence probabilities and continuous DSP physical parameters.

- **Input Features:** 3-channel Log-Mel Spectrogram (Dry, Wet, Delta) computed via T.MelSpectrogram (n_fft=1024, hop=512, n_mels=64) producing shape (B, 3, 64, 173) for 2.0s audio segments.
- **Backbone:** 4-stage 2D Residual Convolutional Network (channels: 3 -> 32 -> 64 -> 128 -> 256) with BatchNorm and LeakyReLU(0.2).
- **Dense Bottleneck:** Concatenated Global Average Pooling (GAP) + Global Max Pooling (GMP) -> 512 dimensions -> Linear -> **256-dimensional dense embedding (emb)**.
- **Multi-Task Heads:**
  1. **Head A (Effect Presence):** Linear(256 -> 7) logits for multi-label classification across all 7 supported effects.
  2. **Head B (Parameter Regression):** Linear(256 -> 29) + Sigmoid yielding continuous normalized parameters strictly in [0.0, 1.0].

Supported Effects & Parameter Ranges (Canonical 29 Continuous Knobs):

Effect Name	Continuous Parameters	Physical Unit	Physical DSP Range	Normalized Range
EQ	frequency gain q	Hz dB dim	[20.0, 20000.0] [-12.0, 12.0] [0.1, 10.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
Compressor	threshold ratio attack release makeup_gain	dB dim ms ms dB	[-60.0, 0.0] [1.0, 20.0] [0.1, 100.0] [10.0, 1000.0] [0.0, 12.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
Delay	delay_time feedback wet_mix dry_mix	ms dim dim dim	[1.0, 2000.0] [0.0, 0.95] [0.0, 1.0] [0.0, 1.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
Reverb	room_size damping wet_mix dry_mix decay	dim dim dim dim s	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.1, 10.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
Distortion	drive tone mix output_gain	dim dim dim dB	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [-12.0, 6.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
Chorus	rate depth delay mix	Hz dim ms dim	[0.1, 10.0] [0.0, 1.0] [1.0, 50.0] [0.0, 1.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
Flanger	rate depth feedback mix	Hz dim dim dim	[0.1, 10.0] [0.0, 1.0] [0.0, 0.95] [0.0, 1.0]	[0.0, 1.0] [0.0, 1.0] [0.0, 1.0] [0.0, 1.0]
STOP	Reserved RL Terminal Action	none	Terminal token for M3 agent	No parameters (rejected by DAPN)

Parameter Mapping Formulation: Executed via common.parameter_mapper.ParameterMapper:

physical = min_phys + clamped_norm * (max_phys - min_phys) (Preserves autograd gradients for all values in [0, 1]).

4. M3 — RL-Based FX-Chain Generation (GRRL-Agent)

Component Status: Planned / Environment Ready. The DAPN differentiable environment, 78/78 unit tests, Multi-Resolution STFT differentiable loss, and parameter mapping are fully operational. The PPO actor-critic loop is currently in active development.

Formal MDP Specification for Chain Synthesis:

- **State Space (s_t):** Concatenation of (1) Acoustic feature embeddings e_audio from M1/M2 representing current rendered audio y_t, dry stem x, and target reference y_ref; (2) Chain step history encoding applied effects; and (3) PTA user taste vector z_taste.
- **Action Space (a_t):** Hybrid discrete-continuous action. Discrete action chooses effect type a_disc in {EQ, Compressor, Delay, Reverb, Distortion, Chorus, Flanger, STOP}. Continuous head outputs normalized parameter knobs a_cont in [0, 1]^K for the selected effect.
- **Multi-Objective Reward:** Total reward balances objective acoustic fidelity, personalized human taste, and chain brevity:

Total Reward: `R_total = R_acoustic + lambda_pta * R_pta - beta * StepPenalty`
- **R_acoustic:** `- MultiResolutionSTFTLoss(y_t, y_ref) computed across 512, 1024, 2048 FFT windows.`
- **R_pta:** `Personalized scalar preference reward signal in [-1.0, 1.0] supplied by PTA.`
- **StepPenalty:** `Regularization penalty encouraging compact, non-redundant effect chains.`

5. PTA-to-M3 Interface Contract & Guidance Modalities

The PTA component can interface with M3 through three mutually compatible guidance modes:

Guidance Mode	Interface Payload / Structure	M3 Consumption & Behavior
1. Reward Guidance (Preference Shaping)	Scalar reward R_pta in [-1.0, 1.0] with optional confidence weight and explanation string.	Injected directly into PPO policy gradient and GAE advantage calculations.
2. Conditioning Vector (Taste Representation)	Dense embedding vector z_taste in R^D (e.g. D=16 or 32, normalized in [-1.0, 1.0]).	Concatenated to policy state input: s_t = [e_audio, e_chain, z_taste].
3. Constraint Masking (Rules & Priors)	JSON dictionary defining prohibited effect types and parameter bounding intervals.	Masks discrete action logits in actor head and clamps continuous parameter actions.

Canonical FX-Chain JSON Interchange Schema (DAPN Contract):

```
{ "session_id": "sess_20261017_001", "sample_id": "sample_00482", "chain": [ { "effect": "EQ", "parameters": {"frequency": 0.450, "gain": 0.650, "q": 0.500} }, { "effect": "Compressor", "parameters": {"threshold": 0.500, "ratio": 0.400, "attack": 0.200, "release": 0.350, "makeup_gain": 0.300} }, { "effect": "Reverb", "parameters": {"room_size": 0.550, "damping": 0.300, "wet_mix": 0.350, "dry_mix": 0.800, "decay": 0.400} } ] }
```

Feedback Event Linking: Human evaluation events occur post-generation (pairwise choice or 5-point rating). Feedback submissions pass back the exact sample_id, linking user ratings directly to the synthesized FX-chain and audio tensor checksum.